

Göther Labs / Evaluated technical improvement

SHA-1 RTL optimization on VTR

Executive result · open 45 nm FPGA proxy

**5.98% lower estimated composite PPA across 64
paired implementations**

Prepared by Göther Labs

Evidence snapshot: 5 August 2026

Academic open-FPGA estimate. Not ASIC signoff, a commercial FPGA result, or manufactured silicon.

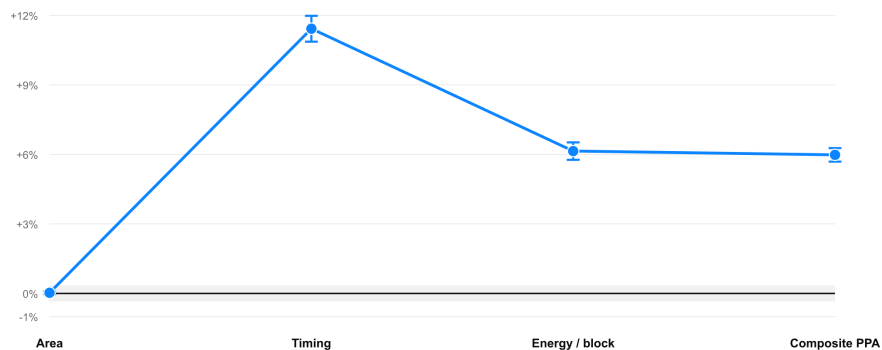
The accepted result

Four cycle-equivalent RTL rewrites improve critical-path delay by 11.43% and workload energy by 6.14%, while total estimated area remains statistically neutral.

Metric	Baseline median	Accepted RTL median	Paired improvement (95% CI)
Total area	16,614,693 MWTa	16,614,693 MWTa	+0.03% (-0.04%, +0.09%) · neutral
Critical path	15.0054 ns	13.28085 ns	+11.43% (+10.87%, +11.98%)
Energy / block	12.3896 nJ	11.6399 nJ	+6.14% (+5.77%, +6.52%)
Composite PPA	1.000000	0.940188	+5.98% (+5.69%, +6.27%)

Certified PPA improvement vs. baseline

64 fixed, paired VPR seeds · lower metric values are better



Paired log-ratio estimate. The gray band is a visual $\pm 0.35\%$ vicinity of no change, not an acceptance threshold.

Paired log-ratio estimates over 64 fixed VPR seeds. Positive values are improvements.

Why it is credible

Only sha.v was editable. NIST conformance and cycle tests passed before PPA; pinned EQY proved sequential equivalence; and a 500-mutation qualification rejected all 482 functionally distinguishable cases. Baseline and accepted RTL were then evaluated under the same 64 fixed implementation seeds.

Claims boundary: this is an open academic FPGA estimate at VTR PTM45, not an ASIC or commercial-FPGA result. SHA-1 is used only as a legacy benchmark.